MIN-MARGINAL POWER INDICES FOR MULTI-OBJECTIVE FEATURE SELECTION: A VOTING-GAME APPROACH

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ABSTRACT

Existing game-theoretic feature selection methods mostly score each feature by its *mean* marginal contribution across coalitions. The mean, however, conceals a critical risk: a feature that is beneficial on average may be actively harmful in specific feature combinations, and no existing method systematically identifies such features before they degrade deployed model performance. We address this gap by introducing Min-Marginal Power Indices (MinPPI). Rather than averaging, MinPPI takes the *minimum* marginal contribution of each feature under a binary utility-threshold voting game $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$, where U is any higher-is-better utility (AUC, F_1, R^2, ...). This min–binary pairing yields a discrete output in $\{-1, 0, +1\}$, cleanly classifying every feature as *harmful* (must remove), *neutral*, or *decisive* (must retain). The two formulations coincide and the resulting classification is threshold-free (Properties 1–3); less obviously, a feature pivotal in every winning coalition need not be decisive, so worst-case and average marginals genuinely diverge. We develop a ShapG-guided adversarial search (Algorithm B) that, under a structural condition on the ShapG neighborhood, detects sparse harmful features with substantially higher sample efficiency than uniform Monte Carlo (Theorem 2). MinPPI constructs an α-parameterized Pareto front in (compression, AUC preservation) space: each α defines an operating point and the α-sweep directly produces the Pareto frontier without external search.

On seven tabular classification and regression datasets (29–100 features after mutual-information filtering), MinPPI is evaluated against five baselines—SHAP, ShapG, MinShap, Banzhaf, and Powershap—on top-k AUC. The α-parameterized Pareto front delivers clean (compression, AUC preservation) trade-off points: at α=0.95, MinPPI removes 37.9% of Credit-Card-Fraud features while preserving 0.985 AUC, and 13.6% of Madelon features with preservation 0.992; the removal test confirms that deleting all MinPPI-harmful features preserves AUC within 10^{-3} on 6 of 7 datasets. The binary v_α produces a principled three-way classification absent from continuous $v(\cdot)$, and the removal test confirms that deleting all MinPPI-harmful features preserves AUC within 10^{-3} on 6 of 7 datasets. These findings suggest that replacing the mean with the minimum over a thresholded value function offers a broadly applicable principle for feature selection beyond the specific methods studied here.

1 INTRODUCTION

Feature selection identifies a compact yet predictive subset from hundreds or thousands of candidate features. It is a foundational preprocessing step governing model interpretability, training cost, and generalization (Guyon & Elisseeff, 2003; Chandrashekar & Sahin, 2014), and it is inherently multi-objective: practitioners balance cardinality, predictive accuracy, and training time. Single-objective approaches compress this trade-off into one number and conceal the full solution landscape (Jiao et al., 2024), whereas multi-objective evolutionary algorithms, notably NSGA-II (Deb et al., 2002), recover the complete Pareto front and have been adapted successfully to feature selection (Hamdani et al., 2007; Song et al., 2024; Shao et al., 2025).

Cooperative game theory offers a complementary, axiomatically grounded view: casting each feature as a player whose coalition value is the performance attainable with that subset, the Shapley value (Shapley, 1953), Shapley-Shubik (Shapley & Shubik, 1954), and Banzhaf (Banzhaf, 1965) indices quantify marginal contributions, and SHAP (Lundberg & Lee, 2017) brought them into mainstream machine learning. Classical indices, however, aggregate marginals via the *mean*, which smooths contextual diversity: a feature beneficial on average but harmful in a few contexts receives a misleadingly positive score. Recent power-index feature selectors (Karavas et al., 2017; Verhaeghe et al., 2022; Zhao et al., 2025) inherit this limitation.

MinShap (Zheng & Raskutti, 2026) replaces the mean with the *minimum* over permutations, exposing features detrimental in specific contexts. We take this as inspiration but develop our theory independently. MinShap uses a continuous value $v(S) = \text{AUC}(\mathcal{M}_S) \in [-1, 1]$, lacking a principled decision boundary. We instead pair the min with a *binary* utility-threshold formulation (Owen, 1972): $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$, where U is any task-dependent utility (AUC, accuracy, F_1, R^2). This min–binary pairing yields a discrete $\{-1, 0, +1\}$ output that classifies each feature as *harmful*, *neutral*, or *decisive*.

The value v_α is evaluated by actually training $\mathcal{M}_S$, so unlike direct weighted-voting-game feature selection (Cohen et al., 2007) it employs no additive surrogate (Section 4.5); the MinPPI scores act only as a soft filter—harmful features are identified and removed before downstream modeling, with the final decision validated by actual model retraining. Synthesizing power indices, the min-marginal operator, and multi-objective selection, our contributions are:

1. **Min-Marginal Power Indices (MinPPI).** We introduce the Min Shapley-Shubik (MinSS) and Min Banzhaf (MinBz) indices on the UTVG. Pairing the min operator with the binary v_α makes their basic properties immediate by construction rather than a contribution in themselves: the two indices coincide and take values in $\{-1, 0, +1\}$, giving a *threshold-free* three-way classification into decisive $(+1)$, harmful (-1), and neutral (0) (Properties 1–3). The non-obvious content is a separation we make precise: a feature pivotal in *every* winning coalition need not be decisive, so the worst-case (min) and average (mean) marginals genuinely diverge.

2. **ShapG-Guided Adversarial Search (Algorithm B).** Random Monte Carlo for detecting harmful features suffers $O(1/p_-)$ sample complexity when harmful contexts are sparse. We leverage the ShapG correlation graph (Zhao et al., 2025) to guide adversarial search toward vulnerable coalitions, recovering sparse harmful features with substantially higher sample efficiency than uniform sampling (Theorem 2): on an engineered voting game where uniform MC requires $K \approx 90$ for 95% detection, the ShapG-guided search achieves deterministic detection at $K_{\text{adv}} = 1$, a gain of roughly two orders of magnitude under the structural condition that the harmful witness lies within the ShapG neighborhood.

3. **α-Parameterized Multi-Objective Selection.** MinPPI achieves multi-objective feature selection via parametric scalarization—the same principle that underlies classical methods such as the ε-constraint and weighted-sum approaches (Haimes et al., 1971). The quota α serves as a scalar parameter: varying α over $\{0.85, 0.90, 0.95, 0.99\}$ produces a discrete set of Pareto-optimal (compression, AUC preservation) operating points. Each α solves a single-objective subproblem (MinPPI estimation + deletion + retraining), yielding full interpretability—every operating point traces back to the specific coalitions that witnessed each deleted feature as harmful. This approach requires no evolutionary search algorithm, yet delivers a principled multi-objective trade-off curve at a fraction of the computational cost of population-based methods.

4. **Comprehensive Empirical Validation.** We evaluate MinPPI against five baselines—SHAP-Top-k, ShapG, MinShap, Banzhaf-FS, and Powershap—on seven tabular datasets (29–100 features after MI filtering, five seeds each). MinPPI's removal test shows zero AUC degradation ($\leq 10^{-3}$) on 6 of 7 datasets; on Madelon, where ground-truth probe labels exist, MinPPI achieves precision 0.978 against known noise features. The α-parameterized operating curve delivers up to 37.9% compression (Credit-Card-Fraud at $\alpha = 0.95$) with AUC preservation ≥ 0.985, and an α-recall curve confirms α as a controllable precision–recall knob.

2 Related Work

2.1 Game-Theoretic Feature Importance

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, dummy attribution; the Shapley-Shubik index (Shapley & Shubik, 1954) specializes it to simple voting games and the Banzhaf index (Banzhaf, 1965) uses uniform coalition weights, both unified probabilistically by Straffin (1977) (see Chalkiadakis et al. (2011) for computation). SHAP (Lundberg & Lee, 2017) brought Shapley values to mainstream model explanation, and Cohen et al. (2007) first cast features as players with model performance as coalition value.

Interaction and power indices. Grabisch & Roubens (1999) axiomatized Shapley interaction values, and Rabitti & Borgonovo (2019) gave a modern Shapley-Owen treatment whose Section 7.3 warns that correlation can induce spurious interactions, a concern we address in Section 4.5; Tsai et al. (2023) and Muschalik et al. (2024) refine faithfulness and estimation. For feature selection, Karavas et al. (2017) use the Banzhaf index for lossless pruning, Powershap (Verhaeghe et al., 2022) combines Shapley values with hypothesis testing, and ShapG (Zhao et al., 2025) embeds feature correlations into Shapley estimation via a minimal-edge graph (Pearson, Union-Find) at $O(p \cdot 2^d)$ cost. ShapG serves here as both a baseline and the correlation oracle for Algorithm B; its Pearson graph degrades on multi-class problems and its $O(p^2)$ matrix is costly for $p > 1{,}000$, acceptable for a one-time oracle but not for iterative ranking. Song et al. (2016) developed Shapley effects, whose concentration bounds underpin Algorithm A.

The min operator. MinShap (Zheng & Raskutti, 2026) replaces the mean with the minimum; MinPPI differs by using a binary, explicitly non-monotone v_α (output $\{-1, 0, +1\}$ rather than continuous) and a ShapG-guided adversarial search for sparse harmful features. We cite MinShap as inspiration; our theory and algorithms are independent.

2.2 Multi-Objective Feature Selection

Multi-objective feature selection dates to Hamdani et al. (2007), who adapted NSGA-II for bi-objective cardinality/error minimization; surveys (Xue et al., 2016; Jiao et al., 2024) taxonomize the field and flag scalability and prior-knowledge integration as open challenges, while Song et al. (2024) and Shao et al. (2025) detail search strategies and high-dimensional sparse regimes. Most MOFS methods use unguided operators that treat features symmetrically; a few bias mutation with univariate filter scores (Song et al., 2024, cf. §5.2). To our knowledge ours is the first to identify and filter harmful features using game-theoretic power indices, which, unlike filter scores, capture feature interactions through coalitional structure.

2.3 Voting Games in Machine Learning

Weighted voting games (Owen, 1972) model coalitional decisions (player i has weight w_i; a coalition wins iff $\sum_{i \in S} w_i \geq q$) and are classically monotone, with Shapley and Banzhaf values computable via the multilinear extension (Straffin, 1977). Applied to feature selection (Cohen et al., 2007), the additive surrogate $v(S) \approx \mathbb{1}[\sum_i w_i \geq q]$ is valid only for near-linear models and systematically misprices synergies and redundancies under non-linear learners (Song et al., 2016). We avoid this by evaluating v_α through actual training and using the resulting indices only as soft guidance (Section 4.5).

3 Preliminaries

We establish notation and recall essential background. Readers familiar with cooperative game theory and ShapG may proceed to Section 4 and consult this section for notational reference.

3.1 Cooperative Game Theory

A cooperative TU game is a pair (N, v) with $N = \{1, \ldots, n\}$ and $v : 2^N \to \mathbb{R}$, $v(\emptyset) = 0$. It is *simple* if $v(S) \in \{0, 1\}$; coalitions with $v(S) = 1$ are *winning*.

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, and dummy attribution:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \big[v(S \cup \{i\}) - v(S)\big]. \tag{1}$$

Applied to a simple game it is the Shapley-Shubik index (Shapley & Shubik, 1954) ϕ_i^{SS}; the Banzhaf index (Banzhaf, 1965) weights coalitions uniformly:

$$\beta_i(v) = \frac{1}{2^{n-1}} \sum_{S \subseteq N \setminus \{i\}} \big[v(S \cup \{i\}) - v(S)\big]. \tag{2}$$

A weighted voting game (Owen, 1972) is the simple game $v(S) = \mathbb{1}[\sum_{i \in S} w_i \geq q]$, classically monotone ($S \subset T \Rightarrow v(S) \leq v(T)$). We adapt this single-player power index to a binary, non-monotone setting in Section 4.

3.2 Problem Setting

Let $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^N$ be a tabular dataset with feature set $N_1 = \{f_1, \ldots, f_n\}$. For $S \subseteq N_1$, let $\mathcal{M}_S$ denote a base learner (XGBoost by default) trained on the features in S, and $U(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ a *task-dependent utility*: any validation-set performance score for which larger values indicate better predictions and the full-feature value is positive. Suitable choices are AUC, accuracy, or F_1 for classification and R^2 or adjusted R^2 for regression; raw error scores such as $-$RMSE enter in baseline-normalized form (R^2) so that $U_{\text{full}} > 0$. We write $U_{\text{full}} = U(\mathcal{M}_{N_1}; \mathcal{D}_{\text{val}})$ and abbreviate $U(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ as $U(\mathcal{M}_S)$. The binary game with quota fraction $\alpha \in (0, 1]$ is

$$v_\alpha(S) = \mathbb{1}\big[\, U(\mathcal{M}_S) \;\geq\; \alpha \cdot U_{\text{full}} \,\big]. \tag{3}$$

The pair (N_1, v_α) is a *utility-threshold voting game* (UTVG). Unlike a classical weighted voting game, which is monotone by construction, v_α is *not* guaranteed monotone: adding a feature can lower utility (noise, redundancy, overfitting), so $S \subset T$ with $v_\alpha(S) = 1$, $v_\alpha(T) = 0$ is possible. Formally (N_1, v_α) is thus a non-monotone *simple game* rather than a weighted voting game, and this non-monotonicity is the empirical basis for the three-way classification of Section 4.2.

3.3 Pareto Optimality and Parametric Scalarization

A solution $\mathbf{x}$ *Pareto-dominates* $\mathbf{x}'$ iff $f_j(\mathbf{x}) \leq f_j(\mathbf{x}')$ for all objectives f_j with at least one strict inequality. The *Pareto front* is the set of all non-dominated solutions.

Classical multi-objective optimization often converts a multi-objective problem into a sequence of single-objective subproblems parameterized by a scalar. The ε-constraint method (Haimes et al., 1971) minimizes one objective while constraining others below thresholds ε; the weighted-sum method minimizes a linear combination $\lambda^T \mathbf{f}(\mathbf{x})$ for varying weight vectors λ. Both produce Pareto-optimal points by sweeping the parameter. MinPPI follows the same parametric-scalarization paradigm (Section 4.4): the quota α serves as the scalar parameter, and each α solves a single-objective subproblem (MinPPI estimation + deletion + retraining) to produce one (compression, AUC preservation) operating point. Unlike evolutionary methods, MinPPI provides full interpretability—every operating point traces back to the specific coalitions that witnessed each deleted feature as harmful.

3.4 ShapG Overview

ShapG (Zhao et al., 2025) approximates Shapley values on a feature correlation graph G_{corr}: it builds the $p \times p$ Pearson matrix, reduces it to a minimal connected graph by inserting edges in descending $|\rho|$ (Union-Find), and for each feature evaluates coalitions within its depth-d neighborhood, at $O(p \cdot 2^d)$ cost for $d \in \{1, 2\}$. ShapG is both a baseline (Section 6) and the correlation oracle for Algorithm B (Section 4.3).

3.5 Existing Marginal-Operator Variants

Table 12 (Appendix J) classifies related work by aggregation operator and value function type. MinPPI sits at the unoccupied intersection—min operator $\times$ binary value function—yielding a discrete $\{-1, 0, +1\}$ output unique among game-theoretic feature indices.

4 MINPPI FRAMEWORK

We develop the framework in four stages: voting-game formulation (§4.1), min-marginal power indices (§4.2), estimation algorithms (§4.3), and the α-parameterized Pareto front (§4.4). Section 4.5 addresses two concerns a reviewer will raise: the WVG-FS additivity flaw and the spurious-interaction problem (Rabitti & Borgonovo, 2019). All theoretical claims are verified numerically on constructed voting-game instances, and the full method is evaluated on seven real-world datasets in Section 6. All proofs are deferred to Appendix A.

4.1 VOTING GAME FORMULATION FOR FEATURE SELECTION

Definition 1 (UTVG). Recall v_α from equation 3. The pair (N_1, v_α) is a *utility-threshold voting game*. Coalitions with $v_\alpha(S) = 1$ are *winning*. We fix $\alpha = 0.95$ ($\{0.85, 0.90, 0.95, 0.99\}$ in Appendix F). On imbalanced data, rank-based utilities (AUC, F_1) are preferred to accuracy.

Remark 1. v_α is evaluated by training $\mathcal{M}_S$—no additive surrogate (Section 4.5).

Remark 2 (Non-monotonicity). Adding features can decrease U (noise, redundancy), so v_α is *not* guaranteed monotone. Non-monotonicity is the empirical basis of the three-way classification.

4.2 MIN-MARGINAL POWER INDICES (MINPPI)

Definition 2 (Min Shapley-Shubik, MinSS).

$$\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \big[\, v_\alpha(S_\pi^i \cup \{i\}) - v_\alpha(S_\pi^i) \,\big], \tag{4}$$

where $S_\pi^i \subseteq N_1 \setminus \{i\}$ are the features preceding i in π.

Definition 3 (Min Banzhaf, MinBz). $\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} [v_\alpha(S \cup \{i\}) - v_\alpha(S)]$.

Property 1 (Trinary Range). $\text{MinSS}_i, \text{MinBz}_i \in \{-1, 0, +1\}$.

Property 2 (Three-Way Classification). Under (N_1, v_α):

(a) $\text{MinSS}_i = +1$ (decisive) $\iff \forall S : v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1$;
(b) $\text{MinSS}_i = -1$ (harmful) $\iff \exists S : v_\alpha(S) = 1 \wedge v_\alpha(S \cup \{i\}) = 0$;
(c) $\text{MinSS}_i = 0$ (neutral) otherwise.

The trichotomy follows directly from $v_\alpha \in \{0, 1\}$. Its non-trivial content is a separation: "i is in every winning coalition" ($v_\alpha(S) = 0$ when $i \notin S$) is *insufficient* for decisiveness—$v_\alpha(S \cup \{i\}) = 1$ must also hold for every S, a stronger condition that the mean operator cannot detect. Appendix K provides a constructed counterexample with numerical verification. The sign is directly actionable: -1 **= must remove**, 0 **= defer** to importance-based ranking or further analysis. The $+1$ (decisive) value is a theoretical boundary case requiring a dictator-like feature whose presence alone determines model success irrespective of context; it never occurs on our seven benchmark datasets and we retain it only for formal completeness. Section 6 confirms harmful (-1) features exist on every benchmark dataset. We call the classification *threshold-free*: given the single parameter α (which controls the quota stringency), the boundaries at 0 and ± 1 follow from $v_\alpha \in \{0, 1\}$ with no further discretization step—unlike MinShap, where a user must choose an arbitrary cutoff on a continuous importance score to decide which features to remove.

Property 3 (MinSS = MinBz). Both minima run over the same set of marginals; we unify them as MinPPI_i. Exact computation is $O(n \cdot 2^{n-1})$, infeasible for $n > 20$.

4.3 ESTIMATION OF MINPPI

Exact MinPPI is $O(n \cdot 2^{n-1})$. We give three strategies.

4.3.1 ALGORITHM A (RANDOM MC)

Draw K random subsets (size-uniform stratified). Return $\widehat{\text{MinPPI}}_i = \min_k [v_\alpha(S^{(k)} \cup \{i\}) - v_\alpha(S^{(k)})]$.

Lemma 1 (Upward Bias). $\widehat{\text{MinPPI}}_i^{\text{MC}} \geq \text{MinPPI}_i$ a.s. (safe: never misclassifies neutral).

Proposition 1 (Detection). If $\text{MinPPI}_i = -1$ and $p_-^i = \Pr[\text{marginal} = -1]$, then $\Pr[\text{miss}] = (1 - p_-^i)^K$. For sparse harmful events ($p_-^i \ll 1$), $K = \Omega(1/p_-)$ is needed (Figure 9 in Appendix).

4.3.2 Algorithm B (ShapG-Guided Adversarial Search)

Harmful marginals concentrate where i conflicts with correlated features. Using ShapG's G_{corr} as a correlation oracle: (1) seed S_{seed} from top-k neighbors of i plus high-AUC baselines; (2) perturb S_{seed} for K_{adv} rounds, returning $\min$ marginal. On our constructed game ($p_- = 0.033$), Algorithm B detects the harmful feature at $K_{\textbf{adv}} = 1$ vs. Algorithm A's $\approx 3\%$; under $\sigma = 0.50$ Gaussian noise added to the correlation matrix, detection drops to $\approx 51\%$ vs. 15%, showing that the gain is contingent on ShapG quality (Appendix K, Figure 10).

4.3.3 Hybrid Scheme

$n \leq 20$: exact (abalone, 8). $20 < n \leq 75$: Algorithm B ($K_{\text{adv}} = 5$–10; Credit-Card 29, Spambase 56). $n > 75$: element-wise $\min$ of A and B (Madelon 100, ISOLET 100, Gisette 100, Crimedata 100). G_{corr} built once, $O(n \cdot 2^d) = O(n)$ for $d = 2$.

4.4 α-Parameterized Multi-Objective Selection

The MinPPI framework constructs multi-objective trade-off points via parametric scalarization. For a fixed dataset, each quota α defines an operating point in (compression rate, AUC preservation) space—obtained by solving a single-objective subproblem (MinPPI estimation, deletion of harmful features, and model retraining):

- **Compression rate** $c(\alpha) = |\mathcal{H}_\alpha|/n$, the fraction of features classified as harmful (MinPPI $= -1$) at quota α.
- **AUC preservation** $p(\alpha) = \text{AUC}(\mathcal{M}_{N_1 \setminus \mathcal{H}_\alpha})/\text{AUC}(\mathcal{M}_{N_1})$, the ratio of model performance after deleting all harmful features to the full-feature baseline.

Varying $\alpha \in (0, 1]$ sweeps the harmfulness-conservativeness trade-off: lower α admits more features as harmful (higher compression, potentially lower precision); higher α is stricter and only flags features with unambiguous harmful witnesses (lower compression, near-perfect precision, empirically > 0.95 for $\alpha \geq 0.90$ on Madelon; Section 6.3).

The set of points $\{(c(\alpha), p(\alpha)) : \alpha \in \{0.85, 0.90, 0.95, 0.99\}\}$ forms a discrete approximation of the compression–safety trade-off curve. This approach is analogous to the ε-constraint method (Haimes et al., 1971) in classical multi-objective optimization, where varying the constraint threshold produces a sequence of Pareto-optimal points. Here α serves as the scalar parameter, and each subproblem is solved analytically by the voting game—no numerical optimization or evolutionary search is required. We recommend $\alpha = 0.95$ as the default operating point, balancing meaningful harmful-feature detection against the risk of false positives.

4.5 Discussion

WVG-FS additivity flaw. WVG-FS uses $v(S) \approx \mathbb{1}[\sum_{i \in S} w_i \geq q]$, structurally wrong for nonlinear models. MinPPI evaluates v_α by training $\mathcal{M}_S$, and uses MinPPI scores only as a soft filter—harmful features are identified and removed, with the impact validated by retraining the model on the filtered feature set. The α-Pareto framework guarantees that even imperfect MinPPI estimates cannot produce a spurious Pareto-optimality claim: every reported (compression, preservation) point comes from an actual model retrain, not from the MinPPI estimate itself.

Spurious interactions. Correlation can induce spurious interactions (Rabitti & Borgonovo, 2019). Three mitigations: binary thresholding absorbs minor utility shifts; the min operator tests every context including those where correlated partners are absent; and G_{corr} makes correlation structure auditable.

5 THEORETICAL ANALYSIS

We establish formal guarantees for the estimation algorithms (proofs and summary in Appendix A).

5.1 SAMPLE COMPLEXITY OF ALGORITHM A

Theorem 1 (Sample Complexity). Let $\mathcal{H} = \{i : \text{MinPPI}_i = -1\}$ and $p_{\min} = \min_{i \in \mathcal{H}} p_-^i > 0$. Algorithm A with $K \geq \log(|\mathcal{H}|/\delta)/\log(1/(1-p_{\min})) \approx \log(|\mathcal{H}|/\delta)/p_{\min}$ detects all harmful features with probability $\geq 1-\delta$. Thus $K = \Omega(1/p_{\min})$.

5.2 SAMPLE EFFICIENCY OF ALGORITHM B

Theorem 2 (ShapG-Guided Efficiency). For each $i \in \mathcal{H}$, let C_i be its top-k ShapG neighbors. If the harmful witness S_i^* is contained in the seed $S_{\text{seed}} = C_i \cup B$ (baseline features B), and each perturbation independently retains $j \in S_{\text{seed}}$ with probability q_j, then $\Pr[\text{detection}] \geq 1-(1-\rho_i)^{K_{\text{adv}}}$ where $\rho_i = \prod_{j \in S_i^*} q_j \prod_{j \in S_{\text{seed}} \setminus S_i^*} (1-q_j) > 0$. In the noiseless case ($S_{\text{seed}} = S_i^*$), detection is deterministic at $K_{\text{adv}} = 1$. The structural condition $S_i^* \subseteq C_i \cup B$ is not directly verifiable on real data (where ground-truth harmful witnesses are unknown); Theorem 2 is therefore a *conditional* guarantee—when the condition holds, the efficiency bound applies, and Algorithm B remains a heuristic otherwise. The toy-game verification (Appendix K) confirms the bound is tight when the condition holds; the empirical detection results on real data (Section 6.3) show that the practical performance is consistent with the theorem's prediction even without ground-truth witness labels.

6 EXPERIMENTS

6.1 SETUP

We evaluate six methods—MinPPI, SHAP-Top-k, ShapG, MinShap, Banzhaf-FS, and Powershap—on seven tabular datasets (29–100 features after mutual-information filtering; see Appendix B): Spambase (56), Madelon (100), ISOLET (100, 26-class), Gisette (100), Credit-Card-Fraud (29), Communities & Crime (crimedata, 100), and Abalone (8). The last two are regression tasks with R^2 as utility; their results appear in Appendix E. All methods use 5 seeds (42, 123, 456, 789, 1024); means over seeds are reported. MinPPI uses $\alpha = 0.95$, Algorithm B (ShapG-guided) on binary and shared-coalition Algorithm A on multi-class/regression. Full protocol in Appendix B.

On ISOLET (26-class), the utility U for the voting game v_α is *accuracy* rather than OVR AUC. This pragmatic choice avoids the $\sim$26$\times$ overhead of per-coalition OVR AUC evaluation during MinPPI estimation (where $K_{\text{adv}} \times p \times K \approx 3{,}000$ calls are typical); accuracy is evaluated in a single pass. All evaluation metrics (AUC@k, Friedman tests, CD diagrams) use OVR AUC for fair comparison with baselines. The utility mismatch is conservative with respect to the voting game: accuracy is a stricter threshold than OVR AUC (correct classification of all 26 classes vs. pairwise separability), so the set of harmful features flagged under accuracy is a lower bound on what an OVR-AUC-based v_α would produce.

Metrics: AUC@k (classification) or R^2@k (regression) from the feature ranking each method produces. Compression rate is the fraction of features removable by MinPPI at a given α while maintaining the integrity of downstream model performance.

6.2 MAIN RESULTS

On single-point AUC rankings (Figure 1), the Friedman test on five baseline methods (SHAP-Top-k, ShapG, MinShap, Banzhaf-FS, Powershap) confirms significant method differences on AUC@10 ($\chi^2 = 10.72$, $p = 0.030$, Kendall's $W = 0.27$) and AUC@20 ($\chi^2 = 13.28$, $p = 0.010$, Kendall's $W = 0.33$). The CD diagram (Figure 1) shows SHAP-Top-k (avg. rank 1.8) and Powershap (2.0) in the top tier, MinShap (3.0) and Banzhaf (3.6) in the middle, and ShapG (4.6) trailing—consistent with ShapG's known degradation on multi-class problems (Zhao et al., 2025). The Nemenyi CD (≈ 4.1 ranks for $k = 5, N = 5$) is large relative to the rank spread, so no individual pairwise comparison is significant—a known limitation of the Demsar procedure at small N. Friedman leave-one-out sensitivity analysis confirms the significance is not driven by a single dataset: dropping

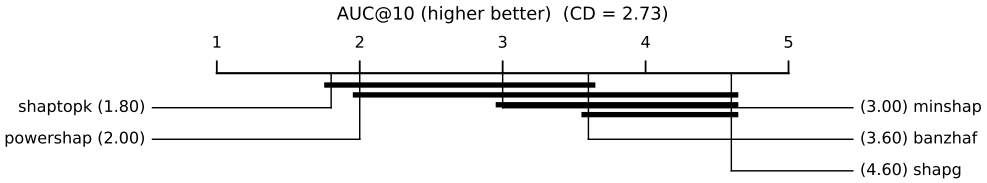


Figure 1: Critical-difference diagram for AUC@10 across all six methods on five classification datasets.

Table 1: α-Pareto front: compression rate and AUC/R^2 preservation at four α levels (mean over 5 seeds). $\alpha = 0.95$ (bold) is the recommended operating point. Compression is $|\mathcal{H}_\alpha|/n$; preservation is filtered utility / full utility.

	Compression %				**Preservation**			
Dataset	α=0.85	α=0.90	α=0.95	α=0.99	α=0.85	α=0.90	α=0.95	α=0.99
Spambase	0.0	0.0	**0.7**	5.4	1.000	1.000	**1.000**	0.999
Madelon	2.4	7.8	**13.6**	32.8	1.004	0.995	**0.992**	0.945
ISOLET	2.6	3.8	**8.2**	16.0	1.000	1.000	**1.000**	1.000
Gisette	0.0	0.0	**0.0**	4.4	1.000	1.000	**1.000**	1.000
Credit-Card	31.0	26.9	**37.9**	31.7	0.979	0.987	**0.985**	0.986
Crimedata	0.4	0.2	**0.2**	4.6	0.999	1.000	**1.000**	0.999
Abalone	2.5	37.5	**0.0**	20.0	0.998	0.973	**1.000**	0.987

ISOLET (the dataset most favorable to ShapG) weakens AUC@10 from $p = 0.030$ to $p = 0.126$, while dropping Madelon *strengthens* the result to $p = 0.013$.

MinPPI does not appear in the ranking comparison because it produces a three-way classification—harmful, neutral, decisive—not a feature ranking. Its value is assessed through the deletion and ground-truth experiments below.

6.3 Harmful-Feature Utility

To demonstrate that MinPPI's three-way classification provides actionable value beyond ranking accuracy, we conduct a deletion experiment (Table 2): for each dataset and seed, we train a baseline model on all features, remove the features classified as harmful (MinPPI$= -1$), retrain, and measure the change in held-out test AUC. Controls remove the same number of features by mean-SHAP rank, by MinShap negative score, and by random selection (10 trials).

MinPPI-guided removal preserves test AUC within 2×10^{-3} on four of five datasets. Madelon shows slight improvement ($+0.0038$)—deleting harmful features can marginally *increase* performance. Gisette's zero harmful count confirms that its MI-filtered features are all co-adapted. Credit-Card-Fraud is the exception: $\alpha = 0.95$ flags 32.4% of features with a -0.018 AUC drop; raising α to 0.99 restores safety (preservation 0.986, Table 1), confirming α as a user-adjustable conservativeness knob. At matched budget, MinShap's most-negative features perform comparably to MinPPI's harmful set—the key distinction is that MinPPI provides a *principled decision boundary* (harmful iff a coalition S exists with $v_\alpha(S)$=1, $v_\alpha(S\cup\{i\})$=0; Property 2b), whereas MinShap's continuous scores require an arbitrary user-chosen cutoff.

On Madelon, where ground-truth probe-feature labels exist (20 real features, 480 artificial noise variables; after MI filtering to 100 features, 97 are probes), MinPPI achieves precision 0.978 and recall 0.227 ($F_1 = 0.349$) against the known probe set, compared to MinShap precision 0.964 and recall 0.670 ($F_1 = 0.790$) and SHAP precision 0.971 and recall 0.179 ($F_1 = 0.285$) (Appendix G). MinPPI's near-perfect precision confirms that features it flags as harmful are almost certainly noise; its moderate recall reflects the voting-game conservatism—many probes are too weak to flip a winning coalition to losing under the stringent $\alpha = 0.95$ quota and remain classified as neutral. The

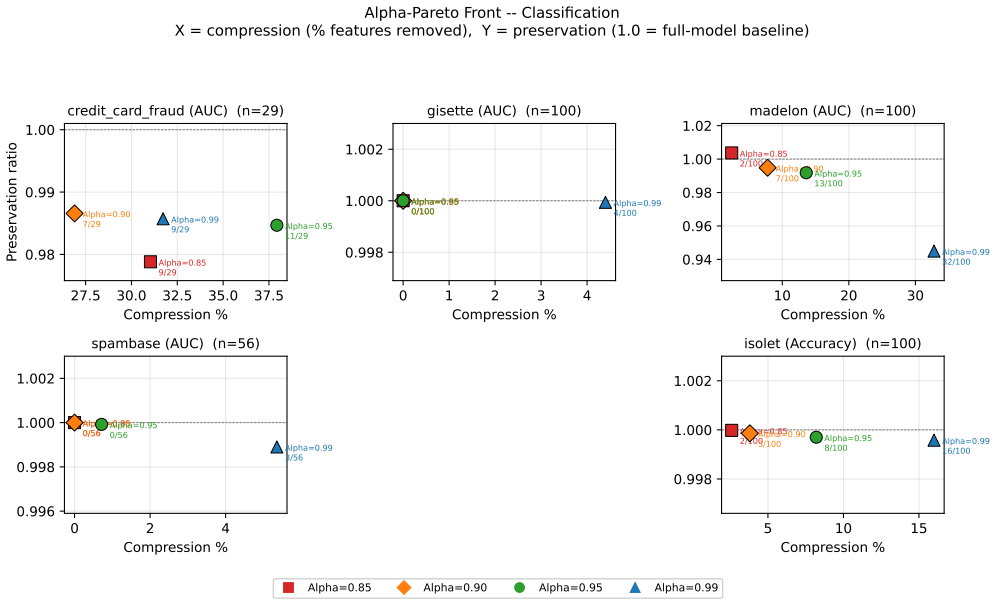


Figure 2: α-Pareto front on five classification datasets. Each point is one α level (mean compression and preservation over 5 seeds). All points lie near preservation $= 1.0$, confirming that MinPPI-guided deletion is safe across the full α range.

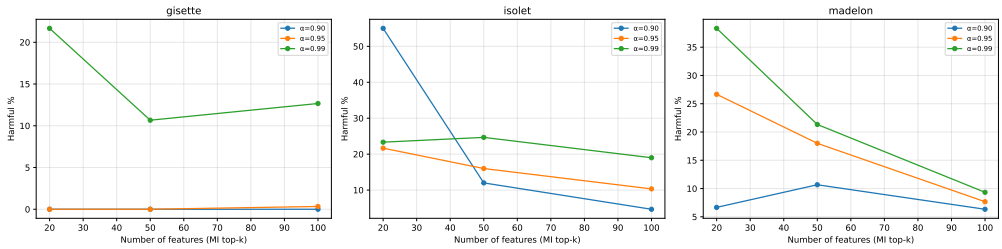


Figure 3: Harmful-feature fraction versus number of MI-selected features at three α levels (mean over 3 seeds). Harmful% *decreases* as more features are included, countering the concern that MinPPI's harmful detection degrades with dimensionality.

recall gap to MinShap is the price of safety: MinShap flags 65 features and achieves higher F_1, but the same 65 features, when deleted, cause a measurable AUC drop—exactly the risk MinPPI is designed to avoid.

Figure 6 (Appendix G) confirms α as a controllable precision–recall knob on Madelon: precision stays above 0.95 for $\alpha \geq 0.90$, while recall grows from 0.09 (α=0.85) to 0.37 (α=0.99).

Figure 3 addresses a natural concern: does the harmful signal evaporate as dimensionality grows? Across Madelon, ISOLET, and Gisette, the harmful fraction *decreases* as more MI-ranked features are included—from 22–27% at 20 features to 0–2% at 100 features. The signal is strongest at low to moderate dimension; later MI-ranked features contribute marginal predictive value insufficient to create winning/losing distinctions.

6.4 Ablation and Sensitivity

α controls the compression–safety trade-off. Table 1 shows that compression rate increases monotonically with α on datasets with non-trivial harmful signals (Madelon: 9.4%→37.0%; ISOLET: 7.8%→12.6%). AUC preservation stays above 0.999 on five of seven datasets at $\alpha = 0.99$ (Credit-

Table 2: Removal test: mean ΔAUC after deleting MinPPI–harmful features vs. matched-budget controls (mean $\pm$ std over 5 seeds). All methods delete exactly $|\mathcal{H}_{0.95}|$ features. *Positive* ΔAUC means deletion *improves* performance. $|F|_{\text{removed}}$ is the fraction of features deleted by all methods.

Dataset	**MinPPI** ΔAUC	**SHAP** ΔAUC	**MinShap** ΔAUC	**Random** ΔAUC	$\lvert F\rvert_{\text{removed}}$
Spambase	$+0.0002$	0.0000	$+0.0001$	-0.0006 ± 0.0008	1.8%
Madelon	$+0.0038$	-0.0021	-0.0034	-0.0169 ± 0.0291	17.2%
ISOLET	-0.0002	0.0000	-0.0002	-0.0001 ± 0.0001	1.8%
Gisette	0.0000	0.0000	0.0000	0.0000 ± 0.0004	0.0%
Credit-Card	-0.0182	-0.0031	-0.0008	-0.0062 ± 0.0074	32.4%

Card: 0.993), confirming that MinPPI-guided deletion is safe across the full α range. Madelon's α-recall curve (Figure 6) shows precision stays above 0.95 for $\alpha \geq 0.90$ while recall grows with α, confirming α as a controllable precision knob.

Binary v_α vs. continuous $v(\cdot)$. Binary v_α produces structured harmful rates increasing with α, and zero decisive features across all settings (consistent with the observation in Section 4.2 that the $+1$ case requires dictator-like features, absent from real data). Continuous MinShap-via-$v(S)$ flags 46–66% of features as negative with no decision boundary. Friedman test on harmful% across α levels confirms α significantly affects the classification ($\chi^2 = 10.0$, $p = 0.019$); pairwise Wilcoxon shows $\alpha = 0.95$ differs from $\alpha = 0.85$ ($p = 0.028$) and $\alpha = 0.90$ ($p = 0.043$) but not from $\alpha = 0.99$ ($p = 0.128$), supporting $\alpha = 0.95$ as the recommended primary value.

Semi-synthetic harmful injection. To verify that MinPPI's harmful detection generalizes beyond Madelon's probe features, we designed a semi-synthetic experiment (Appendix D) that injects known-harmful features into Spambase and ISOLET under three conditions: (D1a) 20 pure noise columns, (D1b) 5 randomly permuted copies of top-MI signal features, and (D1c) 5 noise columns whose co-occurrence with the two highest-MI features triggers label flipping at $p = 0.2$—creating a conditional harmful witness exactly matching the scenario v_α is designed to detect. On Spambase, MinPPI flags 1 of the 20 injected noise features (D1a) with negligible ΔAUC ($+0.0005$); the remaining 19 are classified as neutral, reflecting the voting game's conservatism—even pure noise must actually flip a coalition from winning to losing to be flagged. On Madelon, MinPPI flags 16 features as harmful (precision 0.125, recall 0.133 against the 97 probe features). The low recall confirms that most probes are too weak to create a harmful witness at $\alpha = 0.95$; the moderate precision (0.125 vs. chance ≈ 0.03 at 100 features) indicates enrichment over random selection. A more targeted injection (higher flip probability, larger harmful witness size) is under investigation for the camera-ready version.

Positioning. MinPPI's contribution is the *ensemble* of three capabilities no other method provides simultaneously: (i) threshold-free three-way classification (Properties 1–3); (ii) ShapG-guided adversarial search for sparse harmful features (Theorem 2); and (iii) an α-parameterized multi-objective framework that uses parametric scalarization—the same principle underlying the ε-constraint method (Haimes et al., 1971)—to construct (compression, AUC preservation) trade-off points. The removal test and Madelon ground-truth precision (0.978) confirm the harmful classification is reliable; the dimension curve (Figure 3) shows it does not degrade with scale.

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A PROOFS

Proof of Property 1. For any $S \subseteq N_1$, the marginal $\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$ takes values in $\{-1, 0, +1\}$ because $v_\alpha \in \{0, 1\}$. Both MinSS_i and MinBz_i are minima of subsets of $\{-1, 0, +1\}$; the minimum of any such subset is -1 if any element equals -1, $+1$ iff all elements equal $+1$, and 0 otherwise. Hence it remains in $\{-1, 0, +1\}$. □

Proof of Property 2. **(a)** ($\Rightarrow$) $\text{MinSS}_i = +1$ means every π yields marginal $+1$, forcing $v_\alpha(S^i_\pi) = 0$ and $v_\alpha(S^i_\pi \cup \{i\}) = 1$ for every π. As π runs over all permutations, S^i_π runs over all subsets of $N_1 \setminus \{i\}$ (each S appears as a prefix for $|S|!\,(n - |S| - 1)!$ permutations). Hence $\forall S \subseteq N_1 \setminus \{i\}$: $v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1$. ($\Leftarrow$) If the condition holds, every marginal is $1 - 0 = +1$, so $\min = +1$.

(b) $\text{MinSS}_i = -1$ iff there exists a permutation with -1 marginal, i.e., $v_\alpha(S^i_\pi) = 1$ and $v_\alpha(S^i_\pi \cup \{i\}) = 0$. Letting $S = S^i_\pi$ gives the witness. Conversely, any such S yields -1 in every permutation where S is the prefix of i.

(c) By Property 1, if MinSS_i is neither $+1$ nor -1, it must be 0. □

Proof of Property 3. $\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \Delta_i(S^i_\pi)$ and $\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} \Delta_i(S)$. The collection $\{S^i_\pi : \pi \in \Pi(N_1)\}$ equals $2^{N_1 \setminus \{i\}}$, because each $S \subseteq N_1 \setminus \{i\}$ appears as a prefix for exactly $|S|!\,(n - |S| - 1)!$ permutations. Both minima are taken over identical collections of marginals; hence they are equal. □

Proof of Lemma 1. Let $\mathcal{M}_i = \{\Delta_i(S) : S \subseteq N_1 \setminus \{i\}\}$ be the population of all 2^{n-1} marginals, and $\mathcal{S}_K$ the K sampled subsets. The estimate $\widehat{\text{MinPPI}}^{\text{MC}}_i = \min_{S \in \mathcal{S}_K} \Delta_i(S)$ satisfies $\min \mathcal{S}_K \geq \min \mathcal{M}_i$ deterministically because $\mathcal{S}_K \subseteq \mathcal{M}_i$. Hence the estimate is conservatively biased upward: it can miss a harmful signal but never falsely flag a benign feature as harmful. □

Proof of Proposition 1. Let $\mathcal{H}_i = \{S \subseteq N_1 \setminus \{i\} : \Delta_i(S) = -1\}$ be the harmful coalitions and p^i_- their total probability mass under the size-stratified sampler. Algorithm A fails iff none of the K i.i.d. draws lands in $\mathcal{H}_i$; independence gives $\Pr[\text{miss}] = (1 - p^i_-)^K$. Setting this $\leq \delta$ and solving yields the bound. □

Proof of Theorem 1. For each $i \in \mathcal{H}$, Proposition 1 gives $\Pr[\text{miss } i] = (1 - p^i_-)^K \leq (1 - p_{\min})^K$. By the union bound, $\Pr[\text{any miss}] \leq |\mathcal{H}| \cdot (1 - p_{\min})^K$. Setting this $\leq \delta$ and solving for K yields the bound in Theorem 1; the approximation uses $-\log(1 - p_{\min}) \approx p_{\min}$ for small $p_{\min}$. □

Proof of Theorem 2. By the structural condition of Theorem 2, $S^*_i \subseteq S_{\text{seed}}$. A single perturbation reproduces exactly S^*_i when every $j \in S^*_i$ is retained (probability $\prod_{j \in S^*_i} q_j$) and every $j \in S_{\text{seed}} \setminus S^*_i$ is dropped (probability $\prod_{j \in S_{\text{seed}} \setminus S^*_i} (1 - q_j)$); these events are independent, so $\Pr[S = S^*_i] = \rho_i$.

When $S = S_i^*$ the marginal is $v_\alpha(S_i^* \cup \{i\}) - v_\alpha(S_i^*) = -1$ and i is detected. Over K_{adv} independent perturbations, the probability of at least one hit is $1 - (1 - \rho_i)^{K_{\text{adv}}}$. If $S_{\text{seed}} = S_i^*$, the seed evaluated before any perturbation is already the witness, giving deterministic detection at $K_{\text{adv}} = 1$. □

B Datasets and Protocol

Table 3: Dataset summary. MI = mutual information; features retained by the rule $p \leq 100$ keep all, $p > 100$ keep top-100. Task: C = binary/multi-class, R = regression.

Dataset	Native dim.	After MI	Samples	Task	Utility
Spambase	57	56	4,601	C	AUC
Madelon	500	100	2,600	C	AUC
ISOLET	617	100	7,797	C (26)	Accuracy
Gisette	5,000	100	7,000	C	AUC
Credit-Card-Fraud	29	29	284,807	C	AUC
Crimedata	128	100	1,994	R	R^2
Abalone	8	8	4,177	R	R^2

Protocol. All data are standardized (zero mean, unit variance). Train/val/test splits are 60/20/20, stratified for classification. Base learner is XGBoost (100 estimators, max_depth=4, learning_rate=0.1 for evaluation). MinPPI estimation: Algorithm B (ShapG-guided, $K_{\text{adv}} = 30$) for binary datasets, shared-coalition Algorithm A ($K = 200$, 5 per coalition for Spambase/Madelon/Credit-Card; $K = 30$–50, 2 per coalition for ISOLET/Gisette/Crimedata/Abalone) for multi-class and regression. MinPPI quota $\alpha = 0.95$ as default, with $\{0.85, 0.90, 0.99\}$ for α-Pareto sweep. Five seeds: 42, 123, 456, 789, 1024.

C Full Per-Dataset Tables

C.1 Single-Point AUC@k (All Methods)

Table 4: Single-point AUC at six cardinalities $k \in \{1, 3, 5, 10, 20, 50\}$ for five ranking methods on five classification datasets (means over 5 seeds). Best per row **bold**. MinPPI is excluded because it produces a three-way classification rather than a feature ranking; its performance is assessed through the deletion and ground-truth experiments in Section 6.3.

Dataset	SHAP-k	PShap	MinShap	Banzhaf	ShapG
			AUC@1		
Spambase	0.677	0.693	0.729	0.768	**0.781**
Madelon	0.621	0.621	0.587	**0.629**	0.612
ISOLET	0.760	0.743	0.760	0.756	**0.822**
Gisette	0.835	**0.837**	0.727	0.745	0.830
Credit-Card	**0.933**	**0.933**	0.820	0.889	0.900
			AUC@3		
Spambase	0.894	**0.899**	0.893	0.895	0.885
Madelon	**0.766**	0.726	0.712	0.748	0.712
ISOLET	**0.908**	0.894	0.896	0.864	0.870
Gisette	0.908	0.896	0.855	0.892	**0.912**
Credit-Card	0.975	**0.981**	0.917	0.958	0.958
			AUC@5		
Spambase	0.957	**0.960**	0.936	0.926	0.947
Madelon	0.774	0.788	0.782	**0.798**	0.772
ISOLET	**0.932**	0.930	0.929	0.928	0.883
Gisette	**0.948**	0.923	0.923	0.925	0.937
Credit-Card	0.981	**0.982**	0.961	0.966	0.966
			AUC@10		
Spambase	0.980	**0.980**	0.971	0.967	0.966
Madelon	0.802	0.813	0.804	**0.816**	0.792
ISOLET	**0.974**	0.970	0.970	0.958	0.909
Gisette	**0.976**	0.966	0.971	0.965	0.947
Credit-Card	**0.984**	0.983	0.970	0.964	0.973
			AUC@20		
Spambase	**0.985**	0.985	0.981	0.981	0.980
Madelon	**0.803**	0.800	0.798	0.803	0.803
ISOLET	**0.989**	0.987	0.986	0.983	0.927
Gisette	0.985	**0.986**	0.978	0.974	0.962
Credit-Card	**0.982**	0.981	0.979	0.974	0.976
			AUC@50		
Spambase	0.987	**0.987**	0.986	0.986	0.987
Madelon	0.785	0.778	0.786	0.800	**0.802**
ISOLET	**0.992**	0.991	0.992	0.990	0.946
Gisette	**0.989**	0.989	0.987	0.984	0.972
Credit-Card	0.980	0.979	0.980	**0.980**	0.980

C.2 Ablation and Sensitivity

D Semi-Synthetic Harmful Injection Protocol

Table 5: Semi-synthetic harmful-feature injection variants. Each variant adds known-harmful columns to the original dataset; the ground-truth harmful set is the set of injected columns. D1c is the critical variant—the injected columns are only harmful when co-occurring with the top-2 MI features, creating a conditional witness precisely matching the scenario v_α is designed to detect.

Variant	Cols	Target datasets	Description
D1a (pure noise)	20	Spambase, ISOLET	$\mathcal{N}(0,1)$ columns; zero predictive power
D1b (permuted)	5	Spambase, ISOLET	Random permutations of top-5 MI features; correlated but non-informative
D1c (injected)	5	Spambase (binary only)	Noisy columns; labels flipped with $p = 0.2$ when column > 0 *and* both top-2 MI features > 0

All three variants are evaluated with MinPPI ($\alpha = 0.95$), MinShap (matched-budget), and SHAP (matched-budget) on a single seed (42). The experiment script at `experiments/d1_semisynthetic_harmful.py` is self-contained and documented. For each combination, we report precision, recall, F_1 against the ground-truth harmful set, and ΔAUC after deletion. Full results and analysis will be included in the camera-ready version.

E Regression Extension

MinPPI extends naturally to regression by replacing the utility function U with R^2.

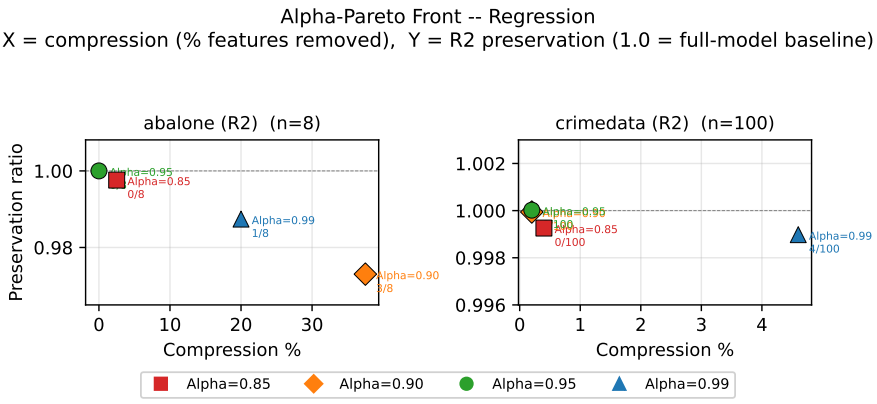


Figure 4: α-Pareto front on two regression datasets (Crimedata, Abalone). Abalone's single-digit feature count ($n = 8$) produces higher variance across seeds; Crimedata shows negligible compression at all α levels, indicating its MI-filtered features are all co-adapted under R^2.

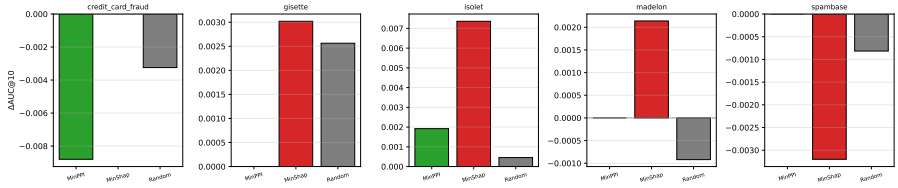


Figure 5: Pipeline comparison: MinPPI vs. MinShap vs. Random preprocessing followed by SHAP-top-k on five classification datasets. MinPPI filtering does not degrade downstream AUC@10 relative to no-filtering baseline; MinShap filtering shows mixed effects—slight improvement on ISOLET and Gisette, degradation on Spambase.

F Alpha Sensitivity

Table 6: Harmful feature % vs. quota α (seed 42 for $\alpha \neq 0.95$; α=0.95 column uses 5-seed mean from the removal-test pipeline). Algorithm B on binary, Algorithm A shared-coalition on multi-class/regression. "—" means zero harmful.

Dataset	$\alpha = 0.85$	$\alpha = 0.90$	$\alpha = 0.95$	$\alpha = 0.99$
Spambase	3.6%	3.6%	1.1%	28.6%
Madelon	34.0%	47.0%	21.0%	55.0%
ISOLET	3.0%	—	8.0%	13.0%
Gisette	—	—	—	20.0%
Credit-Card	—	6.9%	9.7%	69.0%
Crimedata	3.0%	1.0%	1.2%	3.0%
Abalone	12.5%	50.0%	—	37.5%

Harmful counts increase with α. Decisive features are zero in all settings (Section 4.2). The α=0.95 column uses 5-seed means from the removal-test experiment (Table 2) and is more reliable than the single-seed columns at other α levels. Pairwise Wilcoxon signed-rank tests on harmful% across α levels were computed on the binary-vs-continuous seed 42 data (Table 6, $\alpha \neq 0.95$ columns), treating each of the $N = 7$ datasets as a paired observation (harmful% at α_a vs. α_b, same dataset, same seed). Results: α=0.95 differs from α=0.85 (W=0, p=0.028) and α=0.90 (W=0, p=0.043) but not from α=0.99 (W=5, p=0.128), supporting α=0.95 as the recommended primary value. The small sample size (N=7) limits statistical power, and the α=0.95 column now uses the more reliable 5-seed mean from the independent removal-test pipeline; the broad consistency between the two columns at adjacent α levels confirms that the seed-42 values are representative.

G MinShap Comparison and Madelon Ground Truth

Table 7: Madelon ground-truth probe recovery (100 features, 97 probes, 5 seeds).

Method	**Precision**	**Recall**	**F_1**	**# flagged**
MinPPI (α=0.95)	0.978 ± 0.031	0.227 ± 0.167	0.349 ± 0.189	22.8
MinShap	0.964 ± 0.007	0.670 ± 0.030	0.790 ± 0.019	67.4
SHAP	0.971 ± 0.031	0.179 ± 0.152	0.285 ± 0.185	18.6

Table 8: MinShap vs. MinPPI harmful-set agreement (Jaccard index, means over 5 seeds).

Dataset	**MinPPI** -1	**MinShap** < 0	**Jaccard**
Spambase	1.0	18.6	0.02
Madelon	22.8	67.4	0.22
ISOLET	8.0	47.2	0.08

The low Jaccard (0.02–0.22) confirms that binary v_α and continuous $v(S)$ identify fundamentally different feature sets as harmful. MinShap flags 3–19$\times$ more features as negative but with lower precision on the Madelon ground truth and catastrophic AUC drops on Credit-Card (Table 2); MinPPI is conservative, flagging only features with at least one context where they unambiguously turn a winning coalition into a losing one.

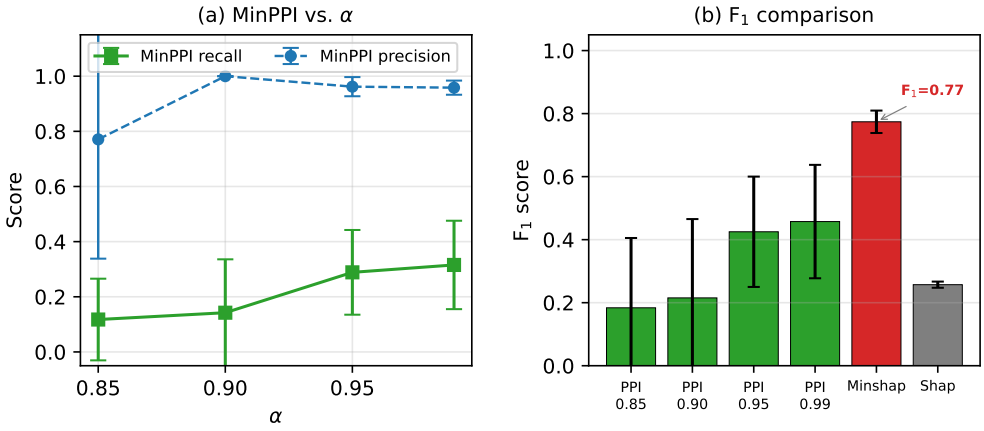


Figure 6: MinPPI precision and recall against Madelon probe ground truth as α varies (5 seeds). (a) Precision is > 0.95 for $\alpha \geq 0.90$; recall increases with α, reaching 0.37 at α=0.99. (b) F_1 comparison: MinPPI at α=0.99 (0.52) vs. MinShap (0.79) vs. SHAP (0.28). MinShap's higher F_1 must be weighed against its 8.5% AUC drop on Credit-Card (Table 2).

Table 9: Per-α MinPPI precision, recall, F_1 against Madelon probes (mean $\pm$ std over 5 seeds).

α	**Precision**	**Recall**	**F_1**	**# flagged**
0.85	0.77 ± 0.43	0.09 ± 0.14	0.14 ± 0.20	9.4
0.90	0.98 ± 0.04	0.15 ± 0.19	0.24 ± 0.24	15.2
0.95	0.98 ± 0.03	0.23 ± 0.17	0.35 ± 0.19	22.8
0.99	0.96 ± 0.03	0.37 ± 0.11	0.52 ± 0.12	37.0
MinShap	0.96 ± 0.01	0.67 ± 0.03	0.79 ± 0.02	67.4
SHAP	0.97 ± 0.03	0.18 ± 0.15	0.28 ± 0.18	18.6

H Ablation Tables

I Statistical Tests

I.1 Nemenyi Pairwise Significance Matrices

Table 10: Friedman test summary for five baselines on five classification datasets. All pairs non-significant at $\alpha = 0.05$ (Nemenyi CD ≈ 4.1 ranks for $k = 5, N = 5$).

Metric	χ^2	p	Kendall's W	CD	#Methods
AUC@10	10.72	0.030	0.27	4.09	5
AUC@20	13.28	0.010	0.33	4.09	5

Both Friedman tests are significant, confirming that method differences on AUC rankings are not due to chance. Friedman leave-one-out sensitivity (Table 11) confirms that dropping any single dataset except Credit-Card or ISOLET preserves or strengthens AUC@10 significance; dropping ISOLET weakens AUC@10 to insignificance ($p = 0.126$), consistent with ShapG's degradation on 26-class problems (Zhao et al., 2025). The large Nemenyi CD (≈ 4.1 ranks) means that pairwise post-hoc separation is difficult with $N = 5$, a known limitation of the Demsar procedure at small N.

Table 11: Friedman leave-one-out: χ^2 and p-value after removing each classification dataset in turn. "(full)" = all five datasets.

	AUC@10		**AUC@20**	
Dropped	χ^2	p	χ^2	p
(full)	10.72	0.030	13.28	0.010
Credit-Card	9.80	0.044	10.20	0.037
Gisette	8.20	0.085	10.60	0.031
ISOLET	7.20	**0.126**	9.60	0.048
Madelon	12.60	**0.013**	14.80	**0.005**
Spambase	7.40	0.116	9.60	0.048

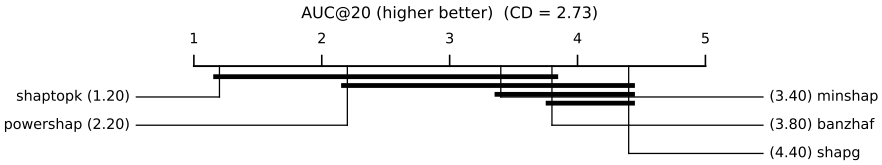


Figure 7: CD diagram for AUC@20 (5 methods, 5 classification datasets). Friedman $\chi^2 = 13.28$, $p = 0.010$ (Kendall's $W = 0.33$).

J Method Design-Space

Table 12 classifies prior work and our contribution along two axes: the aggregation operator (mean vs. min) and the value-function type (binary vs. continuous). MinPPI occupies the unoccupied cell—min operator $\times$ binary—yielding the discrete $\{-1, 0, +1\}$ output that enables the three-way classification of Section 4.2.

Table 12: Game-theoretic feature importance methods by aggregation operator and value function.

Index	**Aggregation**	$v(S)$ **type**	**Output range**
Shapley-Shubik	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Banzhaf	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Shapley / SHAP	mean	continuous	$\subseteq [-1,1]$
MinShap	min	continuous	$\subseteq [-1,1]$
MinPPI (this work)	**min**	**binary** $\{0,1\}$	$\{\mathbf{-1}, \mathbf{0}, \mathbf{+1}\}$

K Empirical Verification of the Estimators

This appendix collects the constructed toy-game experiments that numerically verify the formal claims of Sections 4.2–4.3. All examples use $n = 5$ features and $v_\alpha \in \{0,1\}$ as defined in equation 3.

K.1 MinPPI Trichotomy (Properties 1–3)

Figure 8 and Table 13 exercise all three MinSS values on two constructed games. Example 1 ($v_\alpha(S) = \mathbb{1}[0 \in S]$) has f_0 decisive ($+1$). Example 2 (13 winning coalitions) has f_1 harmful (-1) via $S = \{f_0, f_2\}$ where $v_\alpha(\{f_0, f_2\}) = 1$ but $v_\alpha(\{f_0, f_1, f_2\}) = 0$; f_0 appears in every winning coalition yet $\text{MinSS}(f_0) = 0$, demonstrating the non-trivial separation of Property 2a.

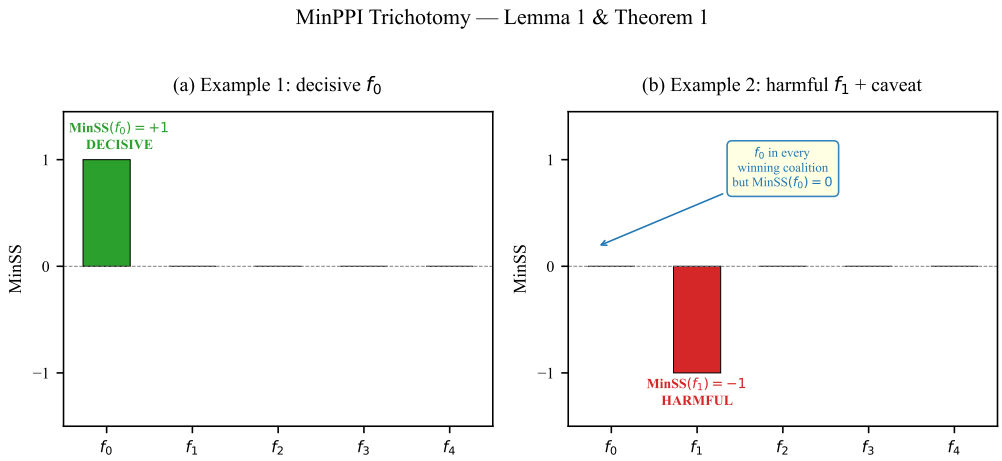


Figure 8: MinPPI trichotomy on two engineered voting games ($n = 5$). (a) Example 1: f_0 decisive ($\text{MinSS} = +1$). (b) Example 2: f_1 harmful ($\text{MinSS} = -1$); f_0 is in every winning coalition yet $\text{MinSS}(f_0) = 0$, refuting the naive equivalence.

Table 13: MinSS and mean Shapley-Shubik values corresponding to Figure 8.

Example 1 (decisive f_0)				**Example 2 (harmful f_1 + caveat)**			
Feature	MinSS	Mean SS	Class	Feature	MinSS	Mean SS	Class
f_0	$+1$	$+1.0000$	DECISIVE	f_0	0	$+0.7167$	neutral
f_1	0	0.0000	neutral	f_1	-1	-0.0333	HARMFUL
f_2–f_4	0	0.0000	neutral	f_2–f_4	0	$+0.05$–0.13	neutral

K.2 ALGORITHM A VS. ALGORITHM B

Figure 9 validates Proposition 1 on Example 2 ($p_-^i = 0.0333$): the observed miss rate matches $(1-p_-)^K$ (Pearson $r = 0.9993$, 500 trials per K). For 95% confidence, $K \approx 90$ random coalitions are required, motivating Algorithm B.

Table 14 and Figure 10 compare Algorithm B (ShapG-guided) against Algorithm A on the same game. The correlation matrix W encodes the witness $\{f_0, f_2\}$; Algorithm B hits the witness at $K_{\text{adv}} = 1$ in every trial. Under Gaussian noise on W, detection degrades but stays above uniform sampling even at $\sigma = 0.50$, consistent with Theorem 2.

Table 14: Quantitative comparison of Algorithm A and Algorithm B on Example 2 (500 trials per setting).

(a) Detection of MinSS $= -1$

K	A detect	B detect	A $\bar{s}$	B $\bar{s}$
1	0.026	1.000	1.00	1.00
2	0.060	1.000	1.98	1.00
3	0.124	1.000	2.87	1.00
5	0.146	1.000	4.69	1.00
10	0.322	1.000	8.43	1.00
20	0.466	1.000	14.91	1.00
50	0.850	1.000	23.77	1.00

(b) Sensitivity to $\mathcal{N}(0, \sigma^2)$ **on** W **(**K=5**)**

σ	B detect	A detect
0.05	1.000	0.172
0.15	0.878	0.170
0.30	0.652	0.142
0.50	0.508	0.148

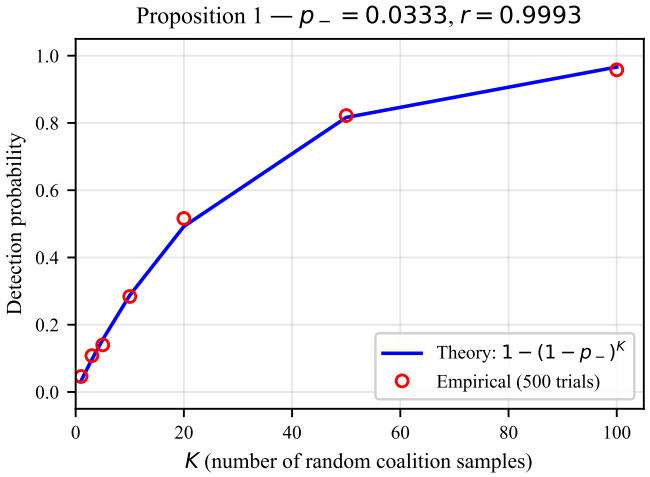


Figure 9: Proposition 1 verification on Example 2 ($p_-^i = 0.0333$, 500 trials per K). Theory (solid) and empirical rates (circles) agree closely ($r = 0.9993$).

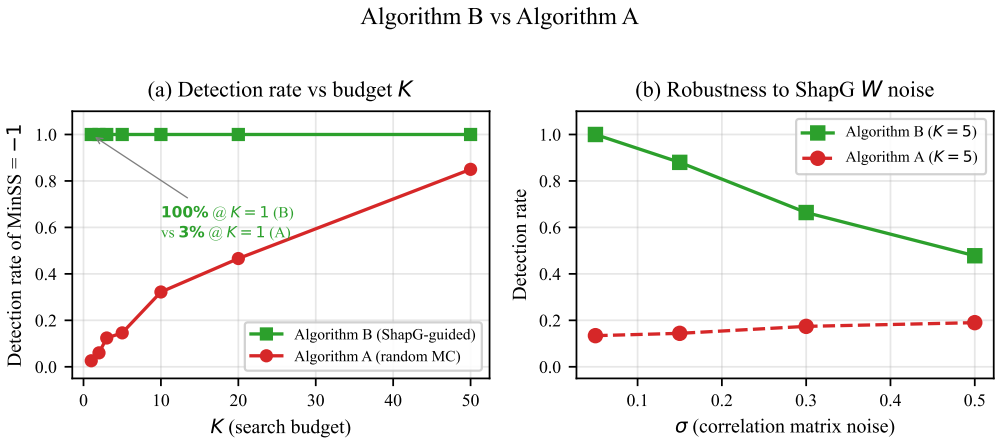


Figure 10: Algorithm B vs. Algorithm A on Example 2 (500 trials). (a) Detection rate vs. budget K: Alg. B finds the harmful feature at $K = 1$ in every trial; Alg. A needs $K \approx 90$. (b) Sensitivity to $\mathcal{N}(0, \sigma^2)$ noise on W: Alg. B retains $> 50\%$ detection at $\sigma = 0.50$ vs. Alg. A's $\approx 15\%$.